1. Problem Statement, Objectives, and Assumptions

Problem Statement

Tracking student performance across multiple examinations manually and determining whether their academic trajectory is improving, declining, or remaining stagnant can be difficult. An automated command-line tool is required to collect exam scores, compute averages, and analyze overall performance trends.

Objectives

- Prompt for the student's name and collect scores across three sequential examinations.
- Calculate the overall average score.
- Determine the performance trajectory by comparing the final exam score with the initial exam score.
- Output a comprehensive performance report showing individual scores, averages, net change, and trend status.

Assumptions

- The evaluation consists of exactly three sequential exams.
- Exam scores are provided as valid numerical inputs.

2. Algorithm, Flowchart, and Pseudocode

Algorithm

1. Prompt the user to enter the student's name.
2.
3. Initialize an empty list called scores.
4.
5. Run a loop 3 times to accept exam scores and append them to scores.
6.
7. Compute the average score by dividing the sum of scores by the total number of exams.
8.
9. Calculate the net change by subtracting the first score (scores[0]) from the last score (scores[-1]).
10.
11. Evaluate the trend:
12.
    - If change > 0, set trend = "Improved".
    -
    - Else if change < 0, set trend = "Declined".
    -
    - Otherwise, set trend = "Unchanged".
    -
13. Print the final performance report.
14.

Flowchart Representation

Pseudocode

3. Python Source Code and Test Cases

Python Source Code

```
        input "Student name: "

for       in range 3
          float input "Enter exam "   str         1    " score: "


          sum           len
                1          0
if          0
          "Improved"elif          0
          "Declined"else
          "Unchanged"

      "\nStudent:"
      "Scores:"
      "Average score:"  round          2
      "Change from first to last exam:"  round          2
      "Performance trend:"
```

(Source code adapted from).

Test Cases & Sample Output

Test Case ID	Scenario Description	User Inputs	Expected Output / Behavior
TC-01	Progressive Improvement	Name: Karan	

Exam 1: 70

Exam 2: 80

Exam 3: 85 | Average: 78.33

Change: +15.0

Trend: Improved. | | **TC-02** | Declining Performance | Name: Neha

Exam 1: 90

Exam 2: 85

Exam 3: 75 | Average: 83.33

Change: -15.0

Trend: Declined. |

Simulated Terminal Output (for TC-01)

4. Conclusion

The student performance trend analysis script provides an effective way to evaluate academic progression across multiple assessments. By calculating averages and comparing baseline scores against final results, it delivers clear, actionable insights into student learning curves.